

Module 3 — Synthesis Quiz

[Question bank](#) · [After exercise lab](#) · [open notes](#) · [two attempts](#)

Module 3, Synthesis quiz

Readings: [Read 0](#), [Read 1](#), [Read 2](#)

Labs: [Demo 1](#), [Demo 2](#), [Exercise](#)

Canvas: Synthesis quiz (Module 3) · two attempts, highest score counts

Full integration for [Project 3](#): five-number summary, boxplot comparisons, scatterplot description, Pearson r , two-way tables and proportions, tidyverse pipeline verbs, ggplot2 calls, and five demo-lab EDA situations.

Graph items: distributed across [Quiz A](#), [Quiz B](#), and this synthesis bank (verbal integration). See also the [Graph Interpretation Pack](#).

Instructions

- Open notes ([methods map](#), [R reference](#)).
- Use the graph-selection table from Read 0 as a guide.
- For R items, match the workflow in [Demo 2](#) and your [Project 3 notebook](#).

Section A — Five-number summary and boxplots (Read 1)

A1. Compute five-number summary

Data (sorted): 2, 5, 7, 8, 10, 12, 15, 18, 22, 30. Compute the five-number summary (min, Q_1 , median, Q_3 , max).

A2. IQR and outlier fence

A dataset has five-number summary min = 2, $Q_1 = 7.5$, median = 11, $Q_3 = 16.5$, max = 30. Compute the IQR and the upper outlier fence ($Q_3 + 1.5 \times \text{IQR}$).

A3. Compare groups from boxplots

Two groups have boxplots: Group A median=25, $Q_1 = 18$, $Q_3 = 32$; Group B median=40, $Q_1 = 35$, $Q_3 = 45$. Which group has higher typical values? Which has more spread?

A4. R code for grouped boxplot

Write the ggplot2 call for boxplots of `body_mass_g` grouped by `species` in the `penguins` dataset.

A5. `group_by` + `summarize` in R

Write the pipeline that computes median and IQR of `body_mass_g` for each `species`.

Answer key — Section A — Five-number summary and boxplots (Read 1)

1. $\text{Min}=2$, $Q_1 = (7 + 8)/2 = 7.5$, $\text{Median}=(10 + 12)/2 = 11$, $Q_3 = (15 + 18)/2 = 16.5$, $\text{Max}=30$.
 2. $\text{IQR} = 16.5 - 7.5 = 9$. Upper fence = $16.5 + 1.5(9) = 30$. The value 30 is on the fence — borderline outlier.
 3. Group B has higher typical values (median 40 vs 25). Group A has more spread ($\text{IQR}=14$ vs $\text{IQR}=10$).
 4. `ggplot(penguins, aes(x = species, y = body_mass_g)) + geom_boxplot()` — optionally add `fill = species` for color.
 5. `penguins %>% group_by(species) %>% summarize(med = median(body_mass_g, na.rm = TRUE), iqr = IQR(body_mass_g, na.rm = TRUE))`
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Section B — Scatterplot description (Read 2 §2)

B1. Describe a scatterplot

A scatterplot of `bill_length_mm` vs `flipper_length_mm` shows points trending upward in a roughly straight band. Describe the association.

B2. Negative association

A scatterplot shows higher values of x paired with lower values of y , with points scattered loosely. Describe the association.

B3. No association

A scatterplot shows no discernible trend — points form a cloud with no slope. Describe the association.

B4. R code for scatterplot

Write the `ggplot2` call for a scatterplot of `bill_length_mm` (x) vs `flipper_length_mm` (y).

B5. Color points by group

How would you modify the scatterplot to color points by `species`?

Answer key — Section B — Scatterplot description (Read 2 §2)

1. Form: linear. Direction: positive. Strength: moderate to strong.
 2. Form: linear (loosely). Direction: negative. Strength: weak.
 3. Form: no clear pattern. Direction: none. Strength: none (weak/no association).
 4. `ggplot(penguins, aes(x = bill_length_mm, y = flipper_length_mm)) + geom_point()`
 5. Add `color = species` to `aes()`: `aes(x = bill_length_mm, y = flipper_length_mm, color = species)`.
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Section C — Two-way tables and proportions (Read 2 §3)

C1. Build a two-way table in R

Write the tidyverse pipeline that counts the number of penguins of each `species` on each `island`.

C2. Compute row proportions

Two-way count table (`species × island`) has row totals: Adelie=152, Chinstrap=68, Gentoo=124. Adelie on Biscoe=44. What is the row proportion of Adelie on Biscoe?

C3. Compute column proportions

Two-way count table (`species × island`): Biscoe column total=168 penguins, and Gentoo on Biscoe=124. What is the column proportion of Gentoo among Biscoe penguins?

C4. Stacked bar chart in R

Write the `ggplot2` call for a 100% stacked bar of `island` (x) with `species` fill.

C5. `pivot_wider` usage

After `penguins %>% count(species, island)`, you want a wide table with `species` as rows and `islands` as columns. Which function converts it?

Answer key — Section C — Two-way tables and proportions (Read 2 §3)

1. `penguins %>% count(species, island)` — or `penguins %>% group_by(species, island) %>% summarize(n = n())`.
 2. $44/152 \approx 0.289$ — about 29% of Adelie penguins are on Biscoe Island.
 3. $124/168 \approx 0.738$ — about 74% of Biscoe penguins are Gentoo.
 4. `ggplot(penguins, aes(x = island, fill = species)) + geom_bar(position = "fill") + labs(y = "Proportion")`
 5. `pivot_wider(names_from = island, values_from = n)` — pivots island names to column headers and fills cells with counts.
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Section D — Choosing the right graph/summary (integration)

D1. Match to variable types

Match each scenario to the correct graph:

- a) Penguin mass by species
- b) Bill length vs flipper length
- c) Species distribution on each island
- d) Distribution of flipper lengths alone

D2. When `group_by` before `summarize`

You have `penguins` and want the mean flipper length for each sex. Write the pipeline.

D3. `filter` before `plot`

You want a scatterplot of only Adelie penguins. What step comes before `ggplot`?

D4. `na.rm` in summaries

Why is `na.rm = TRUE` often needed inside `mean()` or `median()`?

Answer key — Section D — Choosing the right graph/summary (integration)

1. a) Side-by-side boxplots. b) Scatterplot. c) Stacked bar chart. d) Histogram or dot plot.
 2. `penguins %>% group_by(sex) %>% summarize(mean_flip = mean(flipper_length_mm, na.rm = TRUE))`
 3. `filter(species == 'Adelie')` — use `penguins %>% filter(species == 'Adelie') %>% ggplot(...)`.
 4. If any values are NA (missing), R returns NA for the whole mean by default. `na.rm = TRUE` removes missing values before computing the mean.
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Section E — Project 3 workflow (demo lab R patterns)

E1. Load data from URL

You want to load a CSV file from a URL in R. Which function do you use, and where does the URL go?

E2. Recode a variable

Column `sex` has values 1 and 2 (numeric). You want 'male' and 'female'. Which tidyverse function and how?

E3. Handling missing values before plot

Before making a scatterplot of `bill_length_mm` vs `flipper_length_mm`, how do you drop rows with NA in either variable?

Answer key — Section E — Project 3 workflow (demo lab R patterns)

1. `read_csv('https://...')` from the readr package (loaded with `library(tidyverse)`).
 2. `mutate(sex = ifelse(sex == 1, 'male', 'female'))` — or `case_when()` for more than two levels.
 3. `filter(!is.na(bill_length_mm), !is.na(flipper_length_mm))` — or `drop_na(bill_length_mm, flipper_length_mm)`.
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Section F — Graph integration (verbal · no image)

F1. Match graph to types

Match: (a) mass by species (b) bill vs flipper (c) species on island (d) flipper lengths alone.

F2. Wrong graph

Why is a scatterplot wrong for species vs island (both categorical)?

F3. When to add `cor()`

You add `cor()` to quantify the strength of a relationship after making which graph type?

Answer key — Section F — Graph integration (verbal · no image)

1. a) Side-by-side boxplots. b) Scatterplot. c) Stacked bar. d) Histogram or dot plot.
 2. Scatterplots need two numerical axes — use stacked bar or two-way table.
 3. After a scatterplot of two numerical variables (when the form is roughly linear).
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Section G — Pearson r and association (integration)

G1. r sign

$r = -0.73$. Direction?

G2. r vs form

Strong upward curve; $r = 0.2$. What happened?

G3. `cor()` in R

R function for Pearson correlation between x and y ?

Answer key — Section G — Pearson r and association (integration)

1. Negative — higher x tends with lower y .
 2. Form is curved, not linear — r understates the relationship; describe curvature.
 3. `cor(x, y)` (after dropping NA if needed).
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Section H — Tidyverse pipeline verbs (Read 1 · pipeline)

H1. Pipe operator

What is the main purpose of `%>%` (pipe) in the tidyverse?

H2. `filter()`

Which tidyverse function keeps rows that satisfy a logical condition (e.g. `weight > 50`)?

H3. `select()`

Which function chooses columns by name?

H4. `mutate()`

Which function adds a new column while keeping the original number of rows?

H5. `count()`

Which function counts how many rows fall in each level of a categorical variable?

H6. `group_by()` + `summarize()`

What is the role of `group_by(species)` before `summarize(mean_mass = mean(body_mass_g))`?

H7. `summarize()`

After `group_by(island)`, you call `summarize(n = n(), avg = mean(bill_length_mm))`. What do you get?

H8. `glimpse()`

Name two things `glimpse(df)` shows.

Answer key — Section H — Tidyverse pipeline verbs (Read 1 · pipeline)

1. The pipe passes the result of the left side as the first argument to the function on the right — it chains steps so you can read the pipeline top to bottom.
 2. `filter()` — it returns only rows where the condition is `TRUE`.
 3. `select()` — e.g. `select(species, body_mass_g)` returns a data frame with only those columns.
 4. `mutate()` — e.g. `mutate(mass_kg = body_mass_g / 1000)` creates a new column.
 5. `count(variable)` — returns a data frame with one row per level and a column `n` with the count.
 6. `group_by()` splits the data by `species` so `summarize()` returns one summary row per species instead of one row for the whole dataset.
 7. One row per island with two columns: `n` (count of penguins on that island) and `avg` (mean bill length for that island).
 8. Variable names, types (e.g. `<dbl>`, `<chr>`), total rows and columns, and a short preview of values.
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